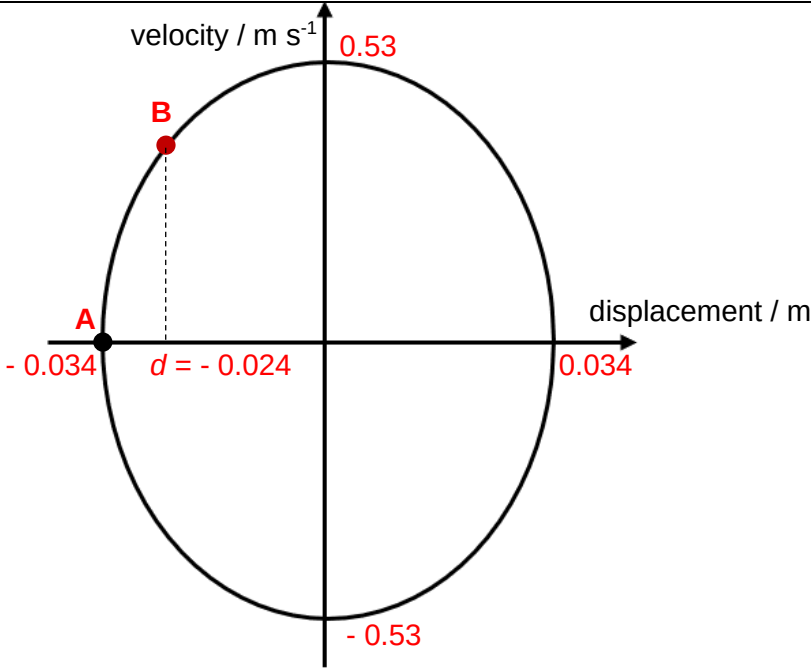


3	(a)	 $v_{\max} = \omega x_0 = 2\pi f x_0 = 2\pi(2.5)(0.034) = 0.534 \text{ m s}^{-1}$ Amplitude and maximum velocity values correctly indicated Point A is labelled correctly		A1 A1
	(b)	$E_T = \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} 1.2 (2\pi(2.5))^2 (0.034)^2$ $= 0.171 \text{ J}$ Or $E_T = \frac{1}{2} m v_{\max}^2 = \frac{1}{2} 1.2 (0.534)^2 = 0.171 \text{ J}$		M1 A1
	(c)	(i)	Any point in second quadrant. (e.c.f if point A is wrongly labelled)	A1
		(ii)	$E_P + E_K = E_T$ When $E_P = E_K = \frac{E_T}{2}$ $\frac{1}{2} m \omega^2 d^2 = \frac{\frac{1}{2} m \omega^2 x_0^2}{2}$ $d = \pm \frac{x_0}{\sqrt{2}} = \pm \frac{0.034}{\sqrt{2}} = \pm 0.024 \text{ m}$ d = - 0.024 m (no answer mark for + 0.024 m) (e.c.f if point B is wrongly labelled as + 0.024 m)	M1 A1

	(d)	$-\frac{x_0}{\sqrt{2}} = -x_0 \cos(\phi) \rightarrow \cos(\phi) = \frac{1}{\sqrt{2}}$ $\phi = \frac{\pi}{4} \text{ rad or } 0.785 \text{ rad}$		M1
			Total:	A1
				9

4	(a)	Two images are just resolved or distinguishable when the central maximum of	B1
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